My research interests lie at the intersection of machine learning, representation learning, computer vision, and healthcare applications. I am particularly interested in how computational methods can extract meaningful structure from complex, high-dimensional data and transform that information into useful scientific or clinical insights. Across my projects, a recurring theme has been understanding how data representation influences the conclusions that can be drawn from a system.

I recently completed a Master's degree in Data Science at Uppsala University. My strongest methodological competencies are in machine learning, deep learning, software development, and data engineering. I have experience developing end-to-end research pipelines, including data collection, preprocessing, annotation, model development, evaluation, and analysis. My work is primarily conducted in Python using frameworks such as PyTorch and the Hugging Face ecosystem.

My master's thesis focuses on multimodal machine learning for harmful content detection on social media platforms. The project combines textual, visual, and contextual information within a unified modelling framework. The work involved designing custom neural architectures, developing representation learning approaches for heterogeneous data sources, evaluating transformer-based models, and analysing how different forms of contextual information influence model performance. A particular focus of the project has been determining which signals provide useful information beyond simple correlations, requiring experimentation under significant uncertainty and limited prior work.

Beyond my thesis, I have worked on projects involving forecasting, anomaly detection, healthcare data systems, and machine learning applications in healthcare. These experiences have strengthened my interest in interdisciplinary research environments where methodological development is closely connected to real-world scientific questions.

Within the context of this PhD position, I am particularly interested in large-scale image analysis, representation learning, and computational modelling of biological systems. The combination of 3D microscopy, shape modelling, and data-driven cell representations offers an opportunity to apply machine learning methods to questions that can directly contribute to biological understanding. I am especially interested in how learned representations can capture meaningful biological variation and support the discovery of patterns that may not be apparent through conventional analytical approaches.